

A study on the detection of trimethylamine gas concentration using copper phthalocyanine

Research Supervisor: Professor Katsutoshi Saeki

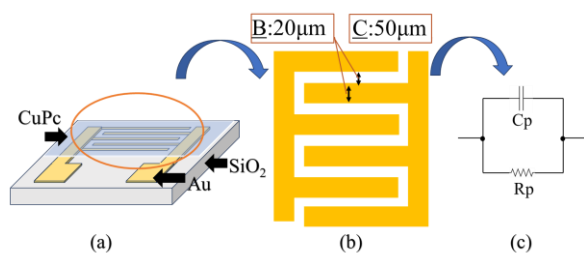
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1. Introduction

In recent years, Trimethylamine (TMA) gas, responsible for foul odors at fish gut processing plants and livestock facilities, has been designated as a harmful air pollutant under the Air Pollution Control Act. The Ministry of the Environment has expressed concern over its potential long-term health effects from continuous exposure. TMA's odor threshold is 0.00044 ppm; visual impairment occurs at 1.2 ppm, and further corneal symptoms—including reduced visual acuity and contrast sensitivity—appear at 2.6 ppm. This study addresses why TMA concentrations above 1.2 ppm could not be measured in previous research. We developed a reductive gas sensor by vacuum-depositing copper phthalocyanine (CuPc) onto comb-shaped electrodes, and examined its suitability for TMA detection.

2. Main Body

Figure 1 presents the schematic, magnified view, and equivalent circuit of the gas sensor. In (a), a glass substrate ($10\text{ mm} \times 12\text{ mm} \times 1.1\text{ mm}$) is shown with gold comb electrodes and a 220 nm vacuum-deposited CuPc film. Adsorption of gas molecules on the CuPc alters the sensor's complex impedance, enabling concentration detection via impedance measurement. The magnified view (b) shows electrode sets with 30 fingers and widths/gaps of $20\text{ }\mu\text{m}$ (B sensor) and $50\text{ }\mu\text{m}$ (C sensor). The equivalent circuit (c) is a parallel RC model reflecting the electrode's resistance and capacitance. Measuring the real part of the impedance (Re) allows gas concentration detection.



(a) Schematic diagram (b) Comb electrode
(c) Equivalent circuit

Figure 1. Gas sensor

Figure 2 presents the block diagram of the TMA gas concentration measurement system. Air is introduced into a permeator PD-1B by a compressor. The air flow and gas concentration are precisely adjusted inside the permeator and then sent into a thermostatic chamber. Inside the chamber, the gas sensor connected to an LCR meter is suspended within a three-neck round-bottom flask. TMA gas is injected from the right side and vented from the left side, while measurements are output to a PC for data processing.

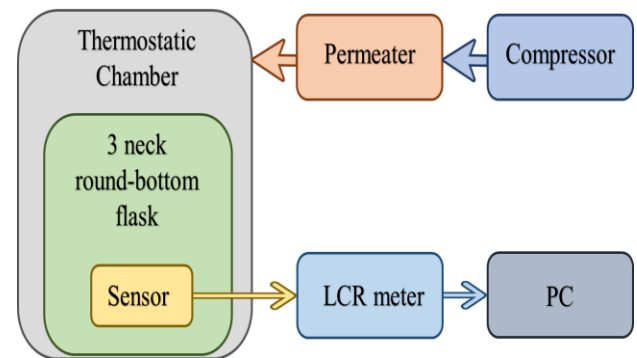


Figure 2. Experimental block diagram

Figure 3 compares the time-dependent real impedance (Re) responses of B and C sensors exposed to TMA concentrations from 0.07 to 2.6 ppm. Measurements in both the thermostatic chamber and permeator were carried out at $35\text{ }^{\circ}\text{C}$. Initial sensor resistances were set to $8\text{ M}\Omega$ (B) and $4\text{ M}\Omega$ (C) to clearly observe changes during repeated exposures. TMA gas was introduced for one minute from 00:30, then ventilated from 01:30. Resistance decreased during exposure and returned to baseline after ventilation. The $30\text{ }\mu\text{m}$ difference in electrode width and spacing between B and C affected their capacitance and impedance, with the C sensor showing smaller capacitance, larger impedance, and less decrease in impedance upon TMA exposure. The C sensor also showed a higher detection limit at high concentrations. The figure demonstrates that higher TMA concentrations cause greater impedance reductions at optimal electrode spacings.

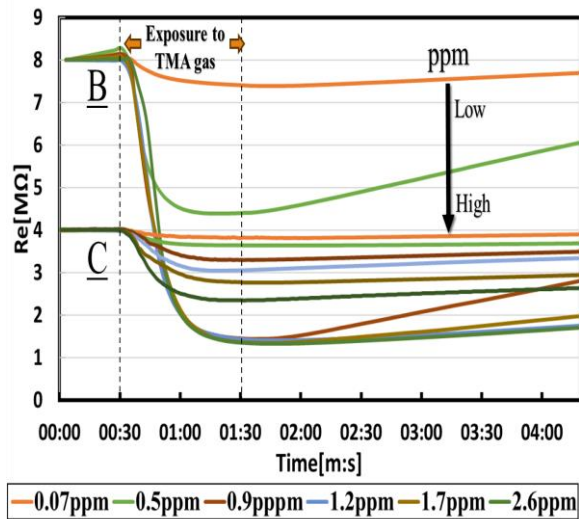


Figure 3. Time-Real Part (Re) of Impedance Characteristics

Figure 4 presents calibration curves for the B sensor, with five repeated measurements at each concentration (0.07–2.6 ppm). The horizontal axis shows TMA concentration, and the vertical axis shows the decrease ratio of the real impedance (ΔRe). Concentrations below 1.2 ppm were measurable, enabling early detection of visual impairment risk.

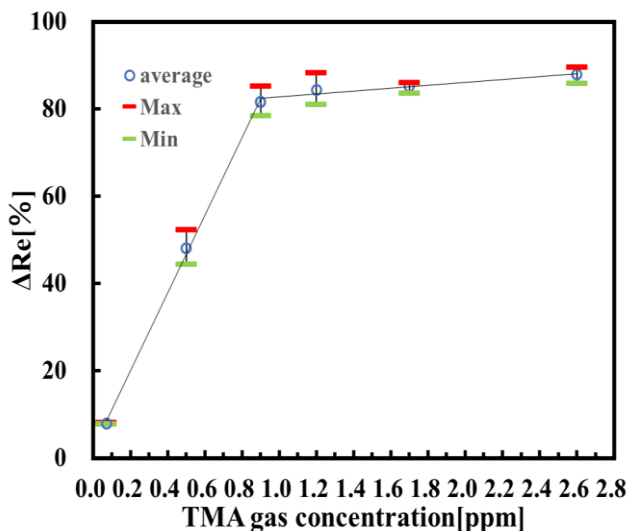


Figure 4. Calibration curve of B sensor (0.07-2.6[ppm])

Figure 5 compares calibration curves of the C sensor (three repeats per concentration) with the B sensor. While B sensor measurements saturated above 1.2 ppm, hindering detection, the C sensor detected concentrations across the full range (0.07–2.6 ppm). These results demonstrate the C sensor's consistent ability to detect unknown TMA levels.

Standard deviations were within 5% for all concentrations, confirming high reproducibility and reliability.

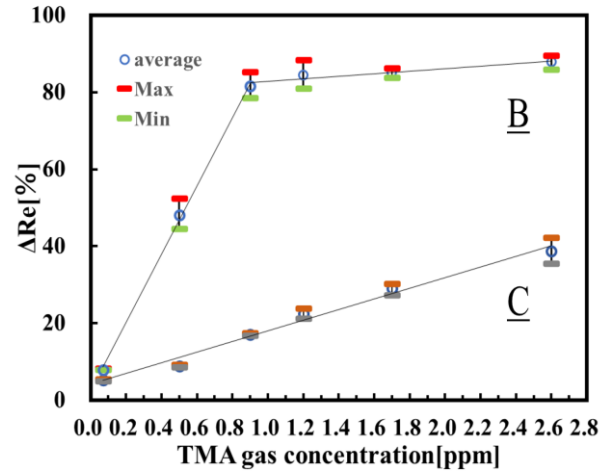


Figure 5. Comparison of Calibration curve (0.07-2.6[ppm])

3. Conclusion

This study investigated why TMA concentrations above 1.2 ppm were difficult to measure in previous research using a CuPc-based reductive gas sensor with comb electrodes. By increasing electrode width and spacing by 30 μm (C sensor), accurate detection of TMA from 0.07 to 2.6 ppm was achieved, enabling early risk detection of visual impairment and corneal symptoms. The sensor also showed high measurement reproducibility. Future work will evaluate selectivity against mixed gases.

4. References

- [1] Jiefan Liu: "Evaluation Method of Triethylamine in Water and Adsorption Properties of Chicken Bone Char for Triethylamine," Tokyo Metropolitan Industrial Technology Research Institute Report, pp.78-79, 2009
- [2] Ministry of the Environment: "Basic Matters Concerning Substances - Trimethylamine," pp.10-13.
- [3] Haruo Kajiya, Katsutoshi Saeki: "Illumination Dependence of Low Concentration Oxidative Gas Sensors Using Copper Phthalocyanine," Proceedings of the 67th Academic Conference of Nihon University College of Science and Technology, M-25, pp.724-723, 2023.
- [4] Shinnosuke Asai, Katsutoshi Saeki, Minoru Saito, Yoshifumi Sekine: "A Study on Odor and Atmospheric Environment Measurement Systems," Nihon University College of Science and Technology Academic Conference Proceedings, M-4, pp.1113-1114, 2012.

【元の日本語論文をどのように翻訳したのか】

Perplexity を使用し，元の日本語論文を読み込ませ，「原文の論文の専門的内容を忠実に反映し，自然で簡潔な英語で表現せよ．また，文章の冗長な部分を整理し，読みやすく短くせよ」と命令した．その後，できた英訳が少し長かったため，原文と同じ 2 枚に収まるように，Perplexity と Chat GPT 両方を使用して，簡潔かつ短い英語に再度翻訳してもらった後，自身の目で再確認をして調整した．

【またなぜその英文にしたのか】

論文投稿基準と英語圏の技術論文の慣例に従い，簡潔かつ明瞭な文章に仕上げた．また，実験の詳細説明は読み手が理解可能であることを重視し，過度な専門用語の省略は避けた英文にした．そして，ガスセンサを gas sensor，数値単位の表記の統一として ppm， μm など日本語の文脈を保持しつつ、英語論文に適した表現に修正をして，研究背景や結果の意味合いが正確かつ明確に伝わるよう工夫した．

【どのように AI を用いたのか】

Perplexity を使用して，まずは元の論文を英訳してもらった．その後，英訳した論文が正しいかを一つの AI だけに頼らずに，「Perplexity と Chat GPT」両方に判断してもらい，誤った表現や回りくどい英語を簡単な英語に直してもらった．

【使った AI の名前やバージョン】

- ・ Perplexity(GPT3.5)を使用(無料版)
- ・ Chat GPT(GPT-3.5)を使用(無料版)

以下が Perplexity に命令した後の最初の出力結果である

添付した論文ファイルを英訳せよ。英訳する際には、論文にふさわしい単語を選び、正確に分かりやすいように英訳せよ。

以下は、添付された論文「銅フタロシアニンを用いたトリメチルアミンガス濃度検知に対する検討」の専門的で正確な英訳です。

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1. Introduction

In recent years, trimethylamine (TMA) gas, which causes foul odors at fish gut processing plants and livestock facilities, has been designated as a harmful air pollutant under the Air Pollution Control Act. The Ministry of the Environment has raised concerns about its long-term adverse effects on human health due to continuous exposure^[1]. The odor threshold for TMA gas is 0.00044 ppm; at 1.2 ppm, visual impairment begins to manifest; at 2.6 ppm, symptoms can also appear on the cornea, including further reductions in visual acuity and contrast sensitivity.

This study investigates the cause for the inability to measure TMA gas concentrations above 1.2 ppm found in previous research. A reductive gas sensor was developed by vacuum-depositing copper phthalocyanine (CuPc) onto comb-shaped electrodes, and its applicability for TMA gas detection was examined.

2. Main Body

Figure 1 shows a schematic diagram, a magnified view, and the equivalent circuit of the gas sensor used. In (a), the schematic depicts a glass substrate (10 mm × 12 mm × 1.1 mm) with comb electrodes made of gold on the substrate. A 220 nm thick CuPc sensitive film is vacuum-deposited onto it. Gas molecules adsorbed on the CuPc alter the complex impedance of the sensor, allowing the detection of gas concentration by measuring impedance variation. The magnified view of the comb electrodes is shown in (b). Each electrode set has 30 fingers with widths and gaps of 20 μm (B sensor) and 50 μm (C sensor). The equivalent circuit depicted in (c) is a parallel RC circuit representing the electrode's resistive and capacitive components. Measuring the real part of the impedance (Re) enables gas concentration detection.

(a) Schematic Diagram (b) Comb Electrode (c) Equivalent Circuit

Figure 1. Gas Sensor

Figure 2 presents the block diagram of the TMA gas concentration measurement system. Air is introduced into a permeator PD-1B by a compressor. The air flow and gas concentration are precisely adjusted inside the permeator and then sent into a thermostatic chamber. Inside the chamber, the gas sensor connected to an LCR meter is suspended within a three-neck round-bottom flask. TMA gas is injected from the right side and vented from the left side, while measurements are output to a PC for data processing.

Figure 2. Experimental Block Diagram

Figure 3 compares the time-dependent real impedance (Re) characteristics of B and C sensors exposed to TMA gas concentrations from 0.07 to 2.6 ppm. Measurements in both the thermostatic chamber and permeator were conducted at a constant temperature of 35°C. The initial sensor resistances were standardized to 8 M Ω for B and 4 M Ω for C sensors to observe sensor changes upon multiple exposures clearly. TMA gas was injected for one minute starting at 00:30, followed by ventilation from 01:30. Resistance decreased upon TMA exposure and returned to baseline with ventilation. The 30 μ m difference in electrode width and spacing between B and C sensors affected their capacitance and impedance characteristics, with the C sensor exhibiting smaller capacitive components and a larger impedance, resulting in smaller decreases in impedance on gas exposure. The C sensor also demonstrated a higher reaction limit for gas uptake at increased concentrations. The figure indicates that higher TMA concentrations lead to larger reductions in impedance at optimal electrode spacings.

Figure 3. Time-Real Part (Re) of Impedance Characteristics

Figure 4 shows calibration curves for the B sensor with five repeated measurements per concentration (0.07–2.6 ppm). The horizontal axis represents TMA concentration, and the vertical axis indicates the impedance real part decrease ratio ΔRe . Concentrations below 1.2 ppm could be measured, which allows early detection of the risk of visual impairment.

Figure 4. Calibration Curve of B Sensor (0.07–2.6 ppm)

Figure 5 compares calibration curves of the C sensor (three repeated measurements per concentration) with those of the B sensor. While previous B sensor measurements saturated above 1.2 ppm making detection difficult, the C sensor enabled concentration detection across the entire range (0.07–2.6 ppm). The results demonstrate the capability of the C sensor to detect unknown TMA concentrations consistently. The standard deviations for all measured concentrations were within 5%, confirming high reproducibility and measurement reliability.

Figure 5. Comparison of Calibration Curves (0.07–2.6 ppm)

3. Conclusion

This study explored the cause behind difficulties in measuring TMA concentrations above 1.2 ppm reported in previous research using a copper phthalocyanine (CuPc)-based reductive gas sensor with comb electrodes. By increasing the electrode width and spacing by 30 μ m (using the C sensor), accurate detection of TMA gas concentrations from 0.07 to 2.6 ppm was achieved. This enables early detection of risks related to visual impairment and corneal symptoms. Furthermore, the sensor demonstrated high reproducibility in measurements. Future work will focus on evaluating selectivity against mixed gases.

4. References

[1] Jiefan Liu: "Evaluation Method of Triethylamine in Water and Adsorption Properties of Chicken Bone Char for Triethylamine," Tokyo Metropolitan Industrial Technology Research Institute Report, pp.78-79, 2009.

Ministry of the Environment: "Basic Matters Concerning Substances - Trimethylamine," pp.10-13.

Haruo Kajiyama, Katsutoshi Saeki: "Illumination Dependence of Low Concentration Oxidative Gas Sensors Using Copper Phthalocyanine," Proceedings of the 67th Academic Conference of Nihon University College of Science and Technology, M-25, pp.724-723, 2023.

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以上が論文の専門的かつ正確な英訳です。必要に応じ、図や詳細な技術説明の英訳も対応可能です。

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